

41.1 Missingness mechanisms

A data set with holes carries two kinds of information: the values that were recorded, and the pattern of which values were recorded at all. The second decides whether any analysis of the first means what it appears to. This section gives the pattern a probability model.

41.1.1 Notation

Write the **complete data** as an array $\mathbf{D}$ with n rows and q columns, whose i th row $\mathbf{d}_{(i)}$ holds the variables of unit i . Response and covariates sit in the same array: which is the response is a question for the analysis, not for the mechanism. Alongside it put the **missingness indicator** array $\mathbf{R}$ of the same shape, with $R_{ij} = 1$ when D_{ij} is recorded; $\mathbf{R}$ is free for this use here, as $\mathbf{H}$ was in Chapters 5 and 10.

The array $\mathbf{R}$ is always completely observed, which is what makes it data. Split $\mathbf{D}$ into recorded entries $\mathbf{D}_{\text{obs}}$ and unrecorded ones $\mathbf{D}_{\text{mis}}$. The analyst holds $(\mathbf{D}_{\text{obs}}, \mathbf{R})$; every model here describes $(\mathbf{D}, \mathbf{R})$ by the factorization

$$f(\mathbf{D}, \mathbf{R} \mid \boldsymbol{\theta}, \boldsymbol{\psi}) = f(\mathbf{D} \mid \boldsymbol{\theta}) f(\mathbf{R} \mid \mathbf{D}, \boldsymbol{\psi}). \quad (41.1.1)$$

Here $\boldsymbol{\theta}$ is the parameter the analyst cares about — a regression coefficient lives in it — and the second factor is the **missingness mechanism**. Conditioning it on the *complete* data is the essential move: it lets the probability that a value is recorded depend on the value itself, which is exactly the case that causes trouble. This is the **selection** factorization; [Section 41.6](#) uses the other one.

41.1.2 Three classes

Definition 41.1.1 — Missing completely at random, at random, not at random

Let $\mathbf{D}$ and $\mathbf{R}$ have the joint distribution [Eq. 41.1.1](#), and fix the realized values $\mathbf{R} = \mathbf{r}$ and $\mathbf{D}_{\text{obs}} = \mathbf{a}$. The mechanism is

- (a) **missing completely at random** (MCAR) if $f(\mathbf{r} \mid \mathbf{D}, \boldsymbol{\psi}) = f(\mathbf{r} \mid \boldsymbol{\psi})$ for every $\mathbf{D}$ and $\boldsymbol{\psi}$: the pattern is independent of the data, recorded or not;
- (b) **missing at random** (MAR) if $f(\mathbf{r} \mid \mathbf{D}, \boldsymbol{\psi})$ takes the same value for every $\mathbf{D}$ whose observed part is $\mathbf{a}$, and every $\boldsymbol{\psi}$, so that it may be written $f(\mathbf{r} \mid \mathbf{a}, \boldsymbol{\psi})$;
- (c) **missing not at random** (MNAR) otherwise.

MCAR implies MAR, and the two buy different things. Under MCAR the recorded units are a random subsample, so *every* analysis of them estimates what it would have estimated with no losses, at the price of a smaller sample. MAR says less: the recorded units are a biased subsample, but the bias is a function of recorded variables, so a method using those variables correctly can undo it.

Warning. The names have misled generations of readers. “Missing at random” does not mean the holes are haphazard; it means they are haphazard *within each configuration of the recorded values* — a strong assumption in modest language. “Missing completely at random” is the haphazard one.

Definition [Definition 41.1.1](#) states MAR at the realized $(\mathbf{r}, \mathbf{a})$; requiring it at every pattern is stronger, and the distinction is not pedantry. A Bayesian, conditioning on the data in hand, needs only the realized version; a frequentist, averaging over data sets that did not occur, needs the stronger one (Seaman, Galati, Jackson and Carlin, 2013). This chapter states the realized version and flags where the stronger one is wanted.

41.1.3 What each class looks like

Example. (A survey that asks about income)

A survey records age x for everyone and income y for those who answer. Three stories. *Completely at random:* a printing fault left the income question off one batch of forms, assembled without regard to anything, so $\Pr(R = 0 \mid x, y)$ is constant. *At random:* younger respondents skip the question more often, and among respondents of a given age, answering has nothing to do with earnings, so $\Pr(R = 0 \mid x, y) = \Pr(R = 0 \mid x)$. *Not at random:* high earners skip the question at every age, so $\Pr(R = 0 \mid x, y)$ depends on y even given x . The three are mutually exclusive descriptions of the world, and [Proposition 41.1.1](#) says the observed data cannot tell them apart.

So “the mechanism” is a property not of a variable but of the pair (variable, recorded variables). If the survey also recorded occupation, and high earners skip only because they hold occupations whose members skip, the third story becomes MAR once occupation joins $\mathbf{D}_{\text{obs}}$. Enlarging the recorded data can turn MNAR into MAR; it can never do the reverse.

Example. (A record with real gaps)

The weekly averages of the continuous Mauna Loa record of atmospheric carbon dioxide run from March 1958 to December 2001 and cover 2284 weeks, of which 59 carry no value — 2.6% of the record, in 22 runs, the longest of 18 consecutive weeks.

They are not spread evenly. Before 1970, 8.6% of weeks are missing; from 1970 on, 0.4%, and a logistic regression of $1 - R_i$ on calendar time multiplies the odds of a gap by 0.192 per decade. The mechanism depends on a variable recorded for every week, so the record is not MCAR.

Whether it is MAR given time is a question about the instrument, not about the numbers. The published rule is that a day contributes only if the analyser ran steadily for at least six hours, and a week with no such day contributes nothing. So the question is whether the analyser’s failure to deliver six steady hours is related to the concentration it would have read — plausibly not, if the failures are electrical or logistical; plausibly so, if unsettled air both disturbs the instrument and moves the reading. Nothing in [Figure 41.1.1](#) can settle that.

```
import numpy as np
import statsmodels.api as sm
```

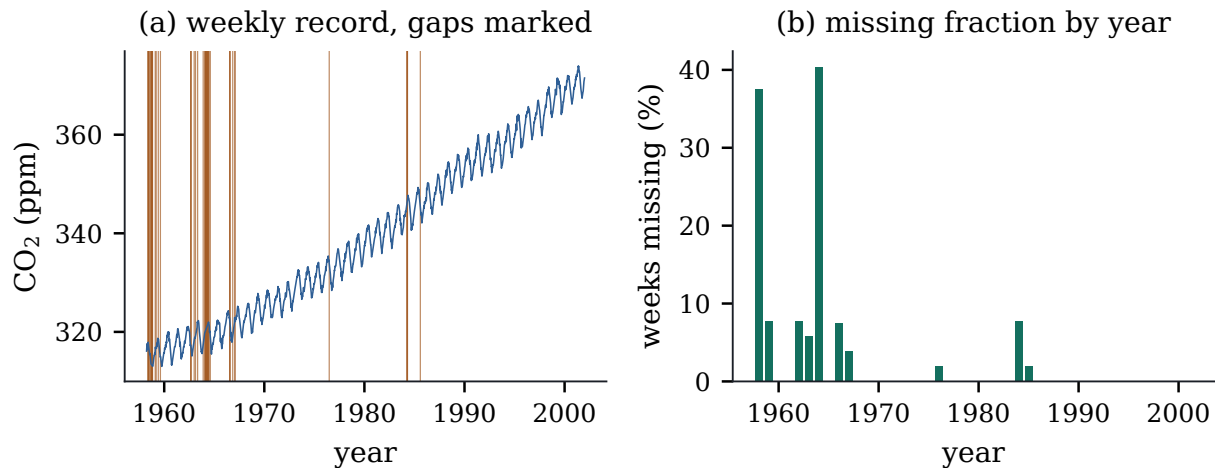


Figure 41.1.1: The weekly Mauna Loa record. (a) The series, with a vertical rule at each missing week. (b) The fraction of weeks missing in each calendar year. The mechanism depends strongly on an observed variable, calendar time, and so is not missing completely at random.

```
co2 = sm.datasets.co2.load_pandas().data["co2"]          # one value per week, some
absent
t = co2.index.year + (co2.index.dayofyear - 0.5) / 365.25
r = (~co2.isna()).to_numpy().astype(float)              # 1 = recorded, 0 = missing
n, n_mis = len(r), int((1 - r).sum())
print(f"{n} weeks, {n_mis} missing ({100 * n_mis / n:.1f}%)")

era = (t >= 1970).astype(float)                          # one observed covariate: is
it 1970 or later?
rate_early = 1 - r[era == 0].mean()
rate_late = 1 - r[era == 1].mean()
print(f"missing before 1970: {100 * rate_early:.1f}%   from 1970 on: {100 *
rate_late:.1f}%")
```

[Run this code online](#)

One feature of **R** matters for computation rather than validity. A pattern is **monotone** when the columns can be ordered so that $R_{ij} = 0$ forces $R_{ik} = 0$ for $k > j$: once a unit drops out it never returns, as in longitudinal dropout. Such patterns are far easier to handle, the joint density factoring into a chain of conditionals each fitted on a nested subsample ([Exercise \(B1\)](#)).

41.1.4 Why the class cannot be read off the data

Proposition 41.1.1 — The observed data do not identify the mechanism

Let $\mathbf{d} = (d_1, d_2)$ be a unit's record, with d_1 always recorded and d_2 recorded exactly when $R = 1$. The distribution of the observed data is determined by the three pieces $f(d_1)$, $\Pr(R = 1 \mid d_1)$ and $f(d_2 \mid d_1, R = 1)$, and determines nothing else. Consequently:

- (a) for every conditional density $g(d_2 \mid d_1)$ there is exactly one joint distribution of $(\mathbf{d}, R)$ with the given observed-data distribution and with $f(d_2 \mid d_1, R = 0) =$

$g(d_2 \mid d_1)$;

- (b) one of these choices, $g = f(\cdot \mid d_1, R = 1)$, is MAR; every other is MNAR; and all assign the observed data the same likelihood.

Proof. The observed data are (d_1, R) , with d_2 when $R = 1$. Their density is $f(d_1) \Pr(R = 0 \mid d_1)$ on $R = 0$ and $f(d_1) \Pr(R = 1 \mid d_1) f(d_2 \mid d_1, R = 1)$ on $R = 1$; only the three listed pieces enter.

- (a) Given the three pieces and a candidate g , define

$$\begin{aligned} f(d_1, d_2, R = 1) &= f(d_1) \Pr(R = 1 \mid d_1) f(d_2 \mid d_1, R = 1), \\ f(d_1, d_2, R = 0) &= f(d_1) \Pr(R = 0 \mid d_1) g(d_2 \mid d_1). \end{aligned}$$

This integrates to one, has the given observed-data distribution, and has conditional density g for d_2 given $(d_1, R = 0)$. Conversely any joint distribution with those properties has these two densities, so it is unique.

- (b) Abbreviate $p_r = \Pr(R = r \mid d_1)$ and $f_1 = f(d_2 \mid d_1, R = 1)$. Under the constructed distribution,

$$\Pr(R = 1 \mid d_1, d_2) = \frac{p_1 f_1}{p_1 f_1 + p_0 g(d_2 \mid d_1)},$$

which is free of d_2 for all d_1 precisely when g equals $f(\cdot \mid d_1, R = 1)$. Every member of the family assigns the observed data the same likelihood by construction, so no statistic computed from those data can prefer one to another. ■

Idea. The missing values are missing. No cleverness recovers the conditional distribution of what was not recorded given what was; a choice of that distribution *is* the assumption, and different choices fit the observed data identically. MAR is one such choice: convenient, often defensible, never verifiable.

Two consequences. A test of MCAR against MAR is possible, that comparison involving recorded variables only, and Little (1988) gives one; a test of MAR against MNAR is not. And a sensitivity analysis is not an optional extra but the only way to report what the data do not determine (Section 41.6).

Finally, the classes of Definition 41.1.1 describe the world; whether a procedure works is a property of the *pair*. Under MCAR nearly every method is valid, including dropping incomplete records. Under MAR with distinct parameters, likelihood and Bayesian inference with a correct model for $\mathbf{D}$ are valid by Theorem 41.3.1, while dropping incomplete records usually is not (Proposition 41.2.1). And under a mechanism depending on a regression's covariates but not on its response, dropping incomplete records is valid *for that regression*, even though the recorded units are a badly biased sample for any other purpose. That third case is the most useful and least well known, and the next section starts there.

41.1.5 Exercises

41.1.5.1 A. Check your understanding

Exercise A1.

Classify each mechanism as MCAR, MAR or MNAR with respect to the recorded variables named.

- (a) A blood assay is run on a random one-third of the samples chosen by a random number generator; age, sex and diagnosis are recorded for all.
- (b) In a weight-loss trial, participants who have gained weight since the last visit stop attending. Weight at every previous visit is recorded.
- (c) A scale reads “over range” and records nothing whenever the true weight exceeds 150 kg.
- (d) A questionnaire item is missing for everyone interviewed by one interviewer, whose identity is recorded.

Solution.

- (a) MCAR. (b) MNAR for the current visit’s weight, since whether it is recorded depends on that weight itself; it would be MAR if attendance depended only on the previous, recorded visits. (c) MNAR, and of an extreme kind: missingness is a deterministic function of the unrecorded value. (d) MAR given the interviewer, and not MCAR; without the interviewer’s identity recorded it would be MNAR *given the recorded variables*, which is what [Definition 41.1.1](#) measures.

Exercise A2.

Show that MCAR implies MAR, and explain why no enlargement of $\mathbf{D}_{\text{obs}}$ can turn a MAR mechanism into an MNAR one.

41.1.5.2 B. Practice**Exercise B1.**

Let $\mathbf{d} = (d_1, \dots, d_q)$ have a monotone pattern, so that the recorded entries are $d_1, \dots, d_K$ for some $K \in \{0, \dots, q\}$. Write the joint density of $(\mathbf{d}, K)$ using the chain rule $f(\mathbf{d}) = \prod_j f(d_j \mid d_1, \dots, d_{j-1})$, and show that MAR is equivalent to $\Pr(K = k \mid \mathbf{d})$ depending on $\mathbf{d}$ only through $d_1, \dots, d_k$.

Solution. Definition [Definition 41.1.1](#) asks that $\Pr(K = k \mid \mathbf{d})$ be the same for all $\mathbf{d}$ agreeing in their first k coordinates, which is the stated condition. Multiplying the chain-rule factorization by it gives

$$f(\mathbf{d}, K = k) = \left\{ \prod_{j \leq k} f(d_j \mid d_{<j}) \right\} \Pr(K = k \mid d_1, \dots, d_k) \left\{ \prod_{j > k} f(d_j \mid d_{<j}) \right\},$$

and integrating out $d_{k+1}, \dots, d_q$ removes the last brace entirely. That is the factorization [Section 41.3](#) exploits.

Exercise B2.

Two variables d_1 (always recorded) and d_2 (recorded when $R = 1$). Propose a statistic with a known null distribution under MCAR and power against “the mechanism depends on d_1 ”, and explain why no statistic can have power against “the mechanism depends on d_2 given d_1 ”.

Exercise B3.

Let $\mathbf{d} = (d_1, d_2)$ be bivariate normal with mean (μ_1, μ_2) and covariance Σ , and let d_2 be recorded with probability $\Phi(a + bd_1)$. Show that the mechanism is MAR, compute $E(d_2 \mid R = 1)$, and confirm that it differs from μ_2 unless $b = 0$ or $\sigma_{12} = 0$. Which class does the mechanism fall in if the probability is instead $\Phi(a + bd_2)$?

Solution. The probability depends only on the recorded d_1 , so the mechanism is MAR. Writing $d_2 = \mu_2 + \beta(d_1 - \mu_1) + e$ with $\beta = \sigma_{12}/\sigma_{11}$ and e independent of d_1 , $E(d_2 \mid R = 1) = \mu_2 + \beta\{E(d_1 \mid R = 1) - \mu_1\}$, and $E(d_1 \mid R = 1) \neq \mu_1$ whenever $b \neq 0$, since the selection probability is monotone in d_1 . With $\Phi(a + bd_2)$ the mechanism depends on the possibly unrecorded d_2 , so it is MNAR for $b \neq 0$.

41.1.5.3 C. Going deeper**Exercise C1.**

Let the units be independent, each with $d_1 \in \{0, 1\}$ always recorded, and let d_2 be recorded with probability p_0 when $d_1 = 0$ and with probability $q(d_2)$ when $d_1 = 1$, where q is not constant. Show that [Definition 41.1.1\(b\)](#) fails at some patterns, but that on any sample in which no unit has $d_1 = 1$ the realized-value condition holds exactly. What does this say about a frequentist argument, which averages over samples that did not occur, compared with a Bayesian one, which conditions on the sample in hand?